

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                               | Marks Available |     |     |       |       |      |
|----------|-----|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 2.       | (a) |      |  | <b>Any two (x1) from</b><br>No hCG {prior to pregnancy/ present before week 4} (1)<br>Rapid rise in levels (if pregnancy is present) (1)<br>Easy to detect/not likely to get false positives (1)                                                                                                                                                                              |                 | 2   |     | 2     |       |      |
|          | (b) | (i)  |  | A (is positive for pregnancy) (1)<br>has a control (line)/ or description of (1)<br>other line shows hCG is present/ ref to antibody sandwich (1)<br>Ignore reference to position of lines                                                                                                                                                                                    |                 |     | 3   | 3     |       |      |
|          |     | (ii) |  | Neither has the control line (1)<br>The unattached enzyme linked antibodies have not bound to the immobilised antibodies (1)                                                                                                                                                                                                                                                  |                 |     | 2   | 2     |       |      |
|          | (c) |      |  | Maintains the corpus luteum (1)<br><br><b>Any two (x1) from</b><br>(if hCG levels fall) then Corpus luteum degenerates/ is not maintained (1)<br>Levels of progesterone would fall/ corpus luteum does not produce enough progesterone (1) Ignore other hormones<br>The {endometrium/uterus lining} {breaks down/ is shed} (1)<br>Embryo would be shed/ cause miscarriage (1) | 3               |     |     | 3     |       |      |
|          | (d) |      |  | Mutations causing {small shape change/ no change in amino acids} (resulting in a functional/normal pregnancy) (1)<br>(Big) differences in {shape/ polypeptides/ protein} would mean {infertility/ miscarriage/ an effect on the pregnancy} (1)                                                                                                                                |                 | 2   |     | 2     |       |      |